

SAURABH KUMAR PANDEY

Specialization in **Artificial Intelligence and Applications**

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SUMMARY

Aspiring Machine Learning Researcher with 3+ years of hands-on experience in fine-tuning and prompting LLMs. Proven ability in designing ML solutions by extracting insights from complex data. Strong expertise in Machine Learning with a research-driven mindset focused on building scalable and intelligent systems.

EDUCATION

- **Indian Institute of Technology, Kharagpur** Kharagpur, India
Masters & Bachelors in Technology - Dual Degree | GPA: 9.10/10.00 2017 - 2022
- **Lucknow Public College** Lucknow, India
Intermediate - 98.25% | High School - 93.20% 2013 - 2016

PUBLICATIONS

- *SMAB: MAB based word Sensitivity Estimation Framework and its Applications in Adversarial Text Generation.* **Saurabh Kumar Pandey**, S. Vashishtha, D. Das, S. Aditya, M. Choudhury. NAACL 2025
- *Reading between the Lines: Can LLMs Help Bridge Cross-Cultural Communication Gaps?* Sougata Saha, **Saurabh Kumar Pandey**, Harshit Gupta, Monojit Choudhury. NAACL 2025
- *Culturally Yours: A Reading Assistant for Cross-Cultural Content.* **Saurabh Kumar Pandey**, Harshit Buddhiraja, Sougata Saha, Monojit Choudhury. COLING 2025
- *Low-resource Counterspeech Generation for Indic Languages: The Case of Bengali and Hindi.* **Saurabh Kumar Pandey**, Mithun Das, S. Sethi, P. Saha, Animesh Mukherjee. EACL 2024
- *Supervised Contrastive Pre-training With Aspect-based Prompts For Aspect Sentiment Triplet Extraction.* Rajdeep Mukherjee, Nithish Kannen, **Saurabh Kumar Pandey**, Pawan Goyal. EMNLP 2023

WORK EXPERIENCE

- **Research Associate II, MBZUAI, Abu Dhabi, UAE** Jul 2024 - Present
Advisor: Professor Monojit Choudhury
Project: Culturally Yours - Cultural Evaluation and Alignment of LLMs
 - Developed an AI cultural reading assistant using LLMs (GPT-4o) to identify CSI spans in review texts
 - Evaluated GPT-4o, Llama-3.1, and Gemma-2 as reading assistants in diverse cultural alignment tasks
 - Studied knowledge topologies in LLMs using logits based spectral analysis of Llama-3.1 and Gemma-2
- **Machine Learning Engineer, V-Labs, Bengaluru, India** Jun 2022 - Jun 2024
Project: DExTr - Information Extraction from Documents using LLMs
 - Implemented a RAG based QnA system for querying complex documents (claims), powered by GPT-3.5
 - Orchestrated an AzureOCR & LLM powered IE module for extracting information from Accord forms
 - Finetuned SLMs using optimized PEFT techniques like LoRA, Q-LoRA, and Prefix Tuning (Acc - 0.62)
 - Improved the accuracy of the extraction pipeline by 15% through suitable document-agnostic promptsProject: Check Fraud - Fraud Detection and Prevention using VLMs and LLMs
 - Extracted information on 10 fields from Bank Checks using Transformer architecture with 87% accuracy
 - Integrated ViT encoder and BART decoder as a pipeline architecture to perform Information Extraction
 - Implemented Siamese based Multi-Scale Residual Fusion Network (F1-0.62) for Handwriting Verification
 - Trained Donut model for extraction of 5 key-value pairs from invoice documents with an accuracy of 93%Project: Damage Assessment - Structural Damage Prediction using VLMs
 - Implemented OneFormer instance segmentation to detect damages (IoU - 0.78) from the rooftop images
 - Employed Open Vocabulary Detection (OWL-ViT) to detect 80+ insurable objects with an mAP of 74%

INTERNSHIPS

- **Machine Learning Intern, mavQ (MTX Group), Hyderabad, India** May 2021 - Jul 2021
 - Developed an object detection model (mAP - 0.7292) using an affirmative ensemble of YOLO-v3 & v4
 - Created Flask APIs and trained 30+ model variations implementing YOLO, RetinaNet & EfficientDet
 - Developed ArcFace loss Face Recognition model (Accuracy - 0.95) on LFW, CFP-FP & AgeDB datasets

TECHNICAL SKILLS

- **Languages:** Python, C/C++, SQL
- **Tools:** Microsoft Azure, AWS, Docker, Git, Slurm, OpenAI, PostgreSQL, pgvector, FAISS, Amazon MTurk
- **Frameworks:** PyTorch, SFTT, LangChain, CrewAI, OpenCV, ScikitLearn, NLTK, TensorFlow, Keras, Flask
- **Publications:** NAACL 2025, COLING 2025, LREC-COLING 2024, EACL 2024, EMNLP 2023, PNAS 2023